Incident Report: Missing Images in NILAM Generated Letter PDFs

Universiti Teknologi MARA — NILAM (Sistem Pengurusan Dokumen Undang-Undang)

Report date	18 August 2026
Prepared for	NILAM production server administrator (root / sudo access required)
Affected host	`nilams.uitm.edu.my`
Affected function	PDF generation for MEU and LPU approval letters (and all other generated letters)
Symptom first seen	On or after 27 July 2026 (reported by users as "it was working before")
Severity	Medium — letters are still produced, but are visually incomplete and not fit to be issued
Action required	Update the operating system CA certificate bundle; restart PHP-FPM. No application code or database change needed.

1. Executive Summary

NILAM generates approval letters as PDF files on demand. Several images embedded in these letters — most visibly the **Jawi (Arabic-script) Hijri month name** in the letter date — are no longer rendering. In their place, the PDF prints the text "`Image not found or type unknown`".

The images are stored on the NILAM server itself and are publicly reachable. The problem is that the PDF rendering library, running *on the server*, must download those images over HTTPS at the moment a PDF is generated — and that download now fails **TLS certificate verification**.

> **Most probable cause:** the TLS certificate for `nilams.uitm.edu.my` was replaced on **27 July 2026** with one issued by a **different certificate authority (SSL.com)**. The server's own CA trust store appears to predate that authority's root certificate, so PHP cannot validate the connection to the site's own hostname.

This is an **operating-system trust-store issue, not an application defect**. The application code that produces these letters has been unchanged since March 2024. Remediation is a routine CA bundle update (Section 6), and it repairs all existing and future letters immediately.

2. Observed Symptom

When a user downloads a letter — for example via `https://nilams.uitm.edu.my/myapplications/approval/8963`, producing `surat_approval_meu.pdf` — the date line renders as:

```
Ruj. Kami: UiTM.100-6/2/3

Tarikh :  1448H  [Image not found or type unknown]  04
```

```
          :  18 Ogos 2026
```

The bracketed placeholder is where the Jawi month name should appear. A second image lower in the letter (the *salam* graphic) is missing in the same way. The university letterhead and the UiTM Hijrah slogan graphic still render correctly.

3. Affected System & How These Letters Are Produced

Understanding the failure requires knowing that **the finished PDF is never stored on the server**. It is rebuilt from scratch on every download:

1. **Draft** — a staff member composes the letter in a rich-text editor. The resulting HTML is saved into the database column `tbl_applications.meu_letter` (or `lpu_letter`). That HTML contains <img> tags holding *absolute URLs*, not the pictures themselves.
2. **Download** — when the letter is requested, the PHP library **dompdf** (v2.0.8, via barryvdh/laravel-dompdf v2.2.0) reads that HTML and must convert each <img> into actual pixels.
3. **Fetch** — to do so, **the web server issues its own outbound HTTPS request** back to `https://nilams.uitm.edu.my/…` to download the image. If that request fails, dompdf substitutes its placeholder text and continues.

> **Important for diagnosis:** checking the image URL in a web browser does **not** test the failing path. The browser uses its own certificate trust store and its own network route. The failure occurs in a request made by PHP, on the server, using the operating system's trust store.

4. Evidence Gathered

4.1 Which images failed

A production-generated PDF (application 8963, reference `100-PUU(32/6/6322)`) was analysed. Its stored letter references three images. Each embedded image in the PDF was identified by matching its pixel dimensions against the live assets:

Image	Source host	Size	Present in PDF?
Letterhead `meu.png`	local disk (relative path, no network)	788 × 138	YES
`hijrah/slogan.png`	`cdn.uitm.edu.my`	314 × 50	YES
`bulan/03.png` (Jawi month)	`nilams.uitm.edu.my`	189 × 55	NO
`salam.png`	`nilams.uitm.edu.my`	1000 × 50	NO

Remote HTTPS image fetching is therefore **working in general** — the `cdn.uitm.edu.my` image downloaded successfully. Only images served from `nilams.uitm.edu.my` failed.

4.2 The two hosts use different certificate authorities

This is the decisive finding. The successful host and the failing host do not share a certificate chain:

	nilams.uitm.edu.my (failing)	cdn.uitm.edu.my (working)

`Leaf issued`	**27 July 2026**	28 July 2025
`Leaf expires`	8 February 2027	24 August 2026
`Intermediate`	SSL.com TLS Issuing RSA CA R1	Sectigo Public Server Authentication CA DV R36
`Root`	**SSL.com TLS RSA Root CA 2022** root created 25 August 2022	Sectigo Public Server Authentication Root R46 → USERTrust RSA Certification Authority root in wide distribution since 2010

The NILAM certificate was replaced on **27 July 2026** and moved to **SSL.com**, whose root certificate dates from August 2022. The CDN host remains on the long-established Sectigo/USERTrust chain, which is present in even very old CA bundles — which is precisely why that one image still works.

4.3 Why PHP is affected when browsers are not

dompdf fetches remote images using PHP's `file_get_contents()` whenever `allow_url_fopen` is enabled; cURL is only used as a fallback (see `vendor/dompdf/dompdf/src/Helpers.php` line 908). `file_get_contents()` validates TLS against the certificate bundle configured for PHP's OpenSSL (`openssl.cafile` / the system store) — a different, and typically much less frequently updated, trust store than the one used by a modern web browser.

4.4 Server platform

The host responds with `Server: Apache/2.4.41 (Ubuntu)`, which corresponds to **Ubuntu 20.04 LTS (released April 2020)**. If the `ca-certificates` package on this host has not been updated since installation, its bundle necessarily predates the SSL.com 2022 root and cannot validate the new certificate.

4.5 Ruled out

Hypothesis	Why it is excluded
Application code change / regression	All 1,663 MEU letters stored since 4 March 2024 contain the identical three image URL patterns. No code change occurred.
Wrong or corrupted URLs in the database	All 8,081 image references across every stored letter resolve to only two hosts, both correct: `nilams.uitm.edu.my` and `cdn.uitm.edu.my`.
Image files missing or web server misconfigured	All image URLs return HTTP 200 with valid PNG content from an external client.
Outbound network blocked / remote images disabled in dompdf	The `cdn.uitm.edu.my` image is fetched successfully in the same PDF render.
WAF blocking non-browser requests by User-Agent	Requests with an empty, browser, and PHP-style User-Agent all returned HTTP 200.

5. Root Cause (to be confirmed on the server)

The server's CA trust store used by PHP does not contain **SSL.com TLS RSA Root CA 2022**. Since the 27 July 2026 certificate replacement, PHP cannot validate a TLS connection to `nilams.uitm.edu.my`, so every image the letter loads from that hostname fails and is replaced by dompdf's placeholder.

Please run the following on the production server to confirm before applying the fix. **All commands are read-only.**

```
#!/bin/bash

# 1. Does PHP's own HTTPS fetch fail for the site's own hostname?

php -r '$d=@file_get_contents("https://nilams.uitm.edu.my/assets/images/bulan/03.png");

        echo $d===false ? "FAIL: ".error_get_last()["message"]."\n" : "OK ".strlen($d)."
bytes\n";'



# 2. Compare against the host that still works

php -r '$d=@file_get_contents("https://cdn.uitm.edu.my/hijrah/slogan.png");

        echo $d===false ? "FAIL: ".error_get_last()["message"]."\n" : "OK ".strlen($d)."
bytes\n";'



# 3. Is the SSL.com 2022 root present in the system bundle?  (0 = confirms the
diagnosis)

grep -c "SSL.com TLS RSA Root CA 2022" /etc/ssl/certs/ca-certificates.crt



# 4. Which bundle is PHP actually using, and how old is the package?

php -r 'print_r(openssl_get_cert_locations());'

dpkg -l ca-certificates | tail -1
```

Expected confirming result: step 1 fails with a message similar to `SSL operation failed … certificate verify failed`, step 2 succeeds, and step 3 returns `0`.

6. Remediation

Update the operating system CA bundle and restart the PHP worker processes so they pick up the new store:

```
sudo apt-get update

sudo apt-get install --reinstall ca-certificates

sudo update-ca-certificates
```

```
sudo systemctl restart php*-fpm

sudo systemctl reload apache2
```

Note on Ubuntu 20.04: this release reached the end of standard support in April 2025. If `apt-get update` reports repository errors, the archive sources may need to be repointed (for example to `old-releases.ubuntu.com`) or an ESM subscription enabled, before the CA package will install.

Alternative if the package cannot be updated: download the SSL.com root certificate, install it locally, and rebuild the bundle:

```
sudo curl -o /usr/local/share/ca-certificates/ssl-com-tls-rsa-root-2022.crt \

     https://ssl.com/repo/certs/SSL.com-TLS-RootCA-RSA-4096-R2.crt

sudo update-ca-certificates

sudo systemctl restart php*-fpm
```

Verify the downloaded certificate's fingerprint against SSL.com's published repository before installing it.

7. Verification After the Fix

1. Re-run step 1 of Section 5. It must now report `OK <n> bytes`.
2. Download a letter that is known to be affected, for example `https://nilams.uitm.edu.my/myapplications/approval/8963` (login required).
3. Confirm the Jawi month name appears in the date line and the *salam* graphic is present, with no "`Image not found or type unknown`" text anywhere in the document.

> Because the PDF is regenerated on every download and is never cached or stored on disk, the fix takes effect **immediately for all existing letters**. No database migration, cache clearing, or content re-entry is required.

8. Risk Assessment

Item	Assessment
Change risk	**Low.** Updating `ca-certificates` is a routine maintenance operation. It adds trusted roots; it does not alter the site's own certificate or its configuration.
Downtime	Brief — only the PHP-FPM restart, typically under two seconds.
Rollback	Not expected to be needed. The previous bundle can be restored from the package cache if required.
Data impact	None. No application data is read or modified.
If left unfixed	All approval letters continue to be issued with missing official graphics, including the Jawi Hijri date required on formal correspondence.

9. Recommendations to Prevent Recurrence

1. **Remove the network dependency from letter generation.** The images are already present on the server's own filesystem, yet the application downloads them over the public internet on every render. Embedding them directly in the letter HTML (as base64 data URIs) would make PDF generation immune to any future certificate, DNS, or network change. This has been tested and confirmed to render correctly in both dompdf and the browser editor. Application-side change; can be scheduled separately.
2. **Add the CA bundle to the certificate renewal checklist.** Whenever the certificate authority for a UiTM host changes, verify that server-to-server HTTPS calls still validate — not only browser access.
3. **Plan an operating system upgrade.** Ubuntu 20.04 is beyond its standard support window; a stale trust store will continue to cause intermittent, hard-to-diagnose failures of this kind.

Appendix A — Technical Reference

Item	Value
PDF renderer	dompdf 2.0.8 via barryvdh/laravel-dompdf 2.2.0
Remote fetch mechanism	`file_get_contents()` — `vendor/dompdf/dompdf/src/Helpers.php:908`
Placeholder text origin	`vendor/dompdf/dompdf/src/Image/Cache.php`
Letter content storage	`tbl_applications.meu_letter` / `tbl_applications.lpu_letter` (type `text`)
PDF storage	None — streamed to the browser, never written to disk
Image assets on server	`public/assets/images/bulan/01.png` … `12.png`, `public/assets/images/salam.png`
Affected letters	All MEU and LPU approval letters — approximately 6,656 stored references to `nilams.uitm.edu.my` images

Appendix B — Certificate Chain Observed

```
$ openssl s_client -connect nilams.uitm.edu.my:443 -servername nilams.uitm.edu.my



 0 s:CN=*.uitm.edu.my

   i:C=US, O=SSL Corporation, CN=SSL.com TLS Issuing RSA CA R1

     notBefore=Jul 27 03:00:51 2026 GMT

     notAfter =Feb  8 09:46:37 2027 GMT

 1 s:C=US, O=SSL Corporation, CN=SSL.com TLS Issuing RSA CA R1

   i:C=US, O=SSL Corporation, CN=SSL.com TLS RSA Root CA 2022
```

```
 2 s:C=US, O=SSL Corporation, CN=SSL.com TLS RSA Root CA 2022

   i:C=US, O=SSL Corporation, CN=SSL.com TLS RSA Root CA 2022   <- root, created Aug
2022
```

Prepared from analysis of a production-generated PDF, the NILAM production database, the application source code, and live TLS inspection of the affected hosts, on 18 August 2026. The root cause in Section 5 is a strongly evidenced hypothesis; the commands provided confirm it definitively before any change is applied.